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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

ME1654-1 – INNOVATION AND DESIGN THINKING

(For CS, EE, CB, BT, CV, AIDS, RI)

Duration: 2 Hours

Max. Marks: 50

Note: Answer Three full questions choosing One full question from each Unit.

Unit – I

	Marks	BT*	CO*	PO*
1. a) What are the features of a good Designed product? Explain with an example of a mobile social media app.	07	L*2	1	2
b) What are the 5 design thinking principles that redefine business and business management? Explain with the help of an industry Which is into college backpack manufacturing.	08	L3	1	2
2. a) Explain 360° approach with suitable example.	07	L1	1	2
b) Define Invention, Innovation and Creativity. Also explain the correlation between them with the help of a case study.	08	L2	1	2

Unit – II

3. a) What are the 7 vital essentials to be contemplated for effective management of distributed team members using agile method in virtual collaboration?	07	L2	2	1
b) What are the five key phases of design thinking in business process modelling? Explain all the five phases taking a incoming crowded college bus in the morning as an example.	08	L2	2	1
4. a) Sketch and explain Agile model of project management taking Water bottle as an example.	07	L2	2	1
b) Explain Business Process Model (BPM) with a help of a flow chart. Also, list the advantages of BPM.	08	L1	2	1

Unit – III

5. There is a 1-acre farm land which is equivalent to 22 numbers of 30X40 Sq ft. site. This land is very close to forest and gets all the serenity and tranquility of nearby forest with some wild animals' habitat like forest giant squirrel, monkeys, deer, wild boar, snakes and some varieties of native birds. The land is placed in a downhill making the rain water flow downwards and pass through the channel located at the center of the land. A corporate working engineer is looking to build a farm house on this land. He has plans of having medicinal plants in one section of the land, some fruit garden in the other section, a small lake to cultivate some fishes in the third section which can in turn serve as the source of water for the land, fourth and fifth sections for dog kennels having 3 different breeds and a cow shed holding around 5 cows, respectively. Empathize with the engineer, define his requirements and ideate the farm land layout for his ease of operation and maintenance with necessary assumptions. He is ready to invest not more than 20 lakhs on this land. Also show the respective price allotment to each section.	20	L3	3	3
6. Considering the first 3 pillars of design thinking, ideate a pair of shoes which can be used for all possible purposes like daily use, jogging, (night and wetland) trekking, formal meeting, biking, family functions and also playing football.	20	L3	3	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
